

BONUS RESOURCE

Excel Formula & Analysis Cheatsheet

Turn raw data into forecasts, charts and insights with one-line Copilot prompts

- 70+ one-line prompts for formulas, cleaning, analysis, forecasting and charts
- The four-part prompt pattern that gets a working formula on the first try
- Agent Mode and Python in Excel: when to reach for them and what to ask
- A verification routine, a troubleshooting table and a printable quick reference

Current as of September 2026, including the retirement of the =COPILLOT worksheet function. Every prompt here is written for the Copilot pane, which is where the capability now lives.

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1 Set the sheet up so Copilot can read it

Seven checks that prevent most failures

Copilot in Excel fails for boring reasons far more often than clever ones. Almost every "it just says it can't do that" comes from this list. Fix these once per workbook and the rest of this pack works.

Check	What to do and why it matters
Licence	Copilot in Excel needs a Microsoft 365 Copilot licence. It is a paid add-on, not part of a standard Personal, Business or Enterprise plan.
File location	The workbook must be saved to OneDrive or SharePoint, with AutoSave on in the desktop app. A local .xlsx is invisible to Copilot.
Format as a table	Select your range and press Ctrl+T. Copilot reads a formatted table far more reliably than a loose range, and can then refer to columns by name.
One header row	A single row of short, unique, text headers. No stacked headers, no blank header cells, no title rows floating above the data.
No merged cells	Merged cells break column structure and are the single most common cause of a wrong or refused answer. Unmerge and use 'Center Across Selection' instead.
Clean data types	Numbers stored as text, mixed date formats and stray spaces produce confidently wrong maths. Section 4 fixes these - do it before you analyse anything.
One table per sheet	If a sheet holds three tables side by side, tell Copilot exactly which range you mean, or split them. Ambiguity costs you more time than the split.

Two-minute habit before any analysis: Ctrl+T the range, check the header row, unmerge everything, and confirm the numeric columns are right-aligned. Right-aligned means Excel sees numbers. Left-aligned means it sees text, and every total you get will be wrong.

2 The four-part prompt pattern

Why "analyse my data" returns nothing you can use

"Analyse my data" is the prompt that gives Copilot a bad name. The prompts in this pack all carry four pieces of information, which is why they return something usable instead of a paragraph of generic observation.

Part	What it does
Location	Name the table, the sheet or the columns. "In the Sales table", "columns B to E", "the Q3 sheet". Never make Copilot guess which data you mean.
Action	One verb. Calculate, compare, rank, forecast, flag, group, clean. Two verbs in one prompt usually gets you one of them done badly.
Rule	The logic in plain English, including the edge case. "Treat blanks as zero", "exclude cancelled orders", "only rows after 1 April".

Part	What it does
Output	Where it goes and in what shape. "As a new column called Margin", "as a PivotTable on a new sheet", "as a formula I can paste into F2", "explain it, do not change the sheet".

THE WEAK VERSION

"Analyse this sales data and tell me what's interesting."

THE VERSION THAT WORKS

"In the Sales table, add a column called Margin % that divides Profit by Revenue, formatted as a percentage with one decimal. Show blanks where Revenue is zero rather than an error, and tell me the formula you used."

Note the last clause. Asking Copilot to state the formula it used turns a black box into something you can check, reuse and explain to whoever asks where the number came from.

3 Formula prompts

Describe the logic, let Copilot pick the function

Copilot is at its strongest here. Describe the logic; let it choose the function. Ask it to add the column directly, or to hand you the formula so you control where it lands.

LOOKUP AND MATCH

"Add a column to the Orders table that pulls the customer's region from the Customers table by matching Customer ID. Show 'Unknown' if there is no match."

"Write a formula that looks up the price for a product and returns the most recent price when a product appears more than once in the price list."

"Match these two lists and give me a column flagging rows that appear in one but not the other, in both directions."

CONDITIONAL LOGIC

"Add a Status column: 'Overdue' if the due date is before today and Paid is blank, 'Due soon' if within seven days, otherwise 'On track'."

"Create a column that bands Revenue into Small, Medium and Large using thresholds of 10,000 and 50,000, and tell me the formula."

"Add a column that flags rows where the value is more than two standard deviations from the column average."

TEXT AND CLEANUP FORMULAS

"Split the Full Name column into First and Last, handling middle names and names with two surnames."

"Write a formula that extracts the invoice number from the Description column, where it always appears as INV followed by six digits."

"Standardise the Company column: trim spaces, fix capitalisation, and strip trailing Ltd, Limited or Inc."

DATES AND PERIODS

"Add a column showing the number of working days between Order Date and Delivery Date, excluding weekends and the dates in my Holidays table."

"Create Month, Quarter and Fiscal Year columns from the Date column. My fiscal year starts in April."

"Add a column showing days since the last order for each customer."

AGGREGATION ACROSS CONDITIONS

"Give me a formula for total revenue for a given region and month, where region is in H1 and month in H2, so I can drag it across a summary grid."

"Count distinct customers per region in the Sales table."

"Add a running total column that resets at the start of each quarter."

EXPLAIN AND REPAIR

"Explain what the formula in cell F2 does, step by step, in plain English."

"This formula returns #N/A for some rows. Tell me why and give me a corrected version that handles the failing cases."

"Rewrite this nested IF as something more readable and tell me what improves."

"Audit my Revenue column for formulas that are inconsistent with the rest of the column."

4 Cleaning messy data

The step that decides whether every later answer is right

Do this before analysis, not after. Every prompt in this section is worth running on a copy of the sheet first - cleaning is destructive, and Copilot's undo is your undo.

"Find every problem in this table that would affect analysis: duplicates, blanks, inconsistent formats, numbers stored as text, mixed date formats and outliers. List them, do not fix anything yet."

"Convert every number stored as text in columns C to F into real numbers, and tell me how many cells changed."

"Standardise the Date column to a single date format and flag any rows where the original value was ambiguous."

"Find duplicate rows based on Email and Order Date, and mark them in a new column rather than deleting them."

"Fill blanks in the Region column with 'Unassigned' and give me a count of how many were filled."

"Trim whitespace and fix inconsistent capitalisation across all text columns."

"Show me every value in the Category column that appears fewer than three times - these are usually typos of the real categories."

"Restructure this data from wide to long: one row per customer per month instead of a column per month."

Ask for the diagnosis before the cure. Run the first prompt, read the list, then instruct the specific fixes you want. A blanket "clean this data" is how a currency column quietly becomes a text column.

5 Analysis and insight

Ask business questions, not spreadsheet questions

This is where you stop doing spreadsheet work and start answering questions. Ask in business language; the pivot, the formula and the sort are Copilot's problem.

FIRST PASS ON UNFAMILIAR DATA

"Describe this dataset: what each column contains, the date range covered, the row count, and anything that looks unreliable."

"What are the three most interesting patterns in this table? For each, tell me the numbers behind it and where you looked."

COMPARISON AND RANKING

"Rank regions by revenue growth between 2025 and 2026, showing both absolute and percentage change."

"Compare this month against the same month last year across every product category, and highlight anything that moved more than 15 per cent either way."

"Which ten customers account for the largest share of revenue, and what percentage of the total is that?"

SEGMENTATION

"Group customers into quartiles by total spend and show average order value and order frequency for each."

"Break revenue down by channel and region in a single table, with row and column totals."

"Show me the revenue split between new and returning customers by month. A customer is new in the month of their first order."

DRIVERS AND DIAGNOSIS

"Revenue fell in Q2. Break the decline down by region, product and channel and tell me where most of the drop came from."

"Which columns are most strongly related to Churn in this table? Report the strength of each relationship and remind me that correlation is not causation."

"Find the rows that are most responsible for the gap between the mean and median of the Order Value column."

TIME AND COHORTS

"Build a cohort table: customers grouped by first-purchase month down the rows, months since first purchase across the columns, retention rate in the cells."

"Show revenue by month with a three-month moving average alongside it."

"Which products show a seasonal pattern, and in which months do they peak?"

6 Forecasting and trend

Specify the method, or you will get a straight line

Copilot will happily forecast anything, which is exactly why you have to be specific about method and assumptions. State the horizon, the basis and what you want it to do with seasonality - otherwise you get a straight line and no warning.

"Forecast monthly revenue for the next six months based on the last 24 months, account for seasonality, and tell me which method you used and why."

"Project this year's total if the last three months' run rate continues, and show the working."

"Give me a best case, base case and worst case for Q4 revenue, and state the assumption behind each in one line."

"Build a simple model: if churn changes by plus or minus two percentage points, what happens to recurring revenue over twelve months? Put the assumptions in their own labelled cells so I can change them."

"Add a trendline calculation to this data and tell me the slope in plain English - how much per month, on average."

"How much of the variation in monthly sales is explained by the trend, and how much is noise?"

"At the current growth rate, when do we cross 1 million in cumulative revenue?"

Treat every forecast as a conversation starter, not an output. Ask which method was used, whether the history is long enough to support it, and what would have to be true for the projection to hold. Microsoft's own guidance is clear that AI output should be reviewed before it reaches anything financial or regulated - and a forecast is exactly that kind of number.

7 Charts, PivotTables and reporting

Say what the chart must prove, not which chart to draw

Say what the chart must show, not which chart to build. Copilot picks the type; you judge whether it made the point.

CHARTS

"Chart monthly revenue for the last two years so the seasonal pattern is obvious, and label the highest and lowest months."

"Show the contribution of each product category to total revenue, ordered largest to smallest."

"Plot budget against actual by month with the variance shown underneath."

"This chart is hard to read. Suggest a clearer way to show the same point and rebuild it."

PIVOTTABLES

"Create a PivotTable on a new sheet with Region in rows, Quarter in columns and Revenue as the value, showing percentage of row total."

"Add a calculated field to this PivotTable showing margin percentage."

"Rebuild this PivotTable to answer: which sales rep improved the most between halves?"

FORMATTING AND PRESENTATION

"Apply conditional formatting to highlight the top 10 per cent of values in green and the bottom 10 per cent in red, with a legend cell explaining the rule."

"Format this sheet for printing on one page wide: freeze the header row, set sensible column widths and repeat headers on every page."

"Write a two-paragraph summary of what this sheet shows for someone who will not open the file."

"Turn this analysis into six bullet points for a slide, leading with the number that matters most."

8 Agent Mode and Python in Excel

When the task is a sequence rather than a question

The Copilot pane answers questions. Agent Mode does jobs - it plans several steps and executes them in the workbook. Use it when the task has a sequence, not when you want one formula.

Use the pane when You want one formula, one chart, one answer, or an explanation. Fast, cheap to verify, easy to undo.

Use Agent Mode when The task is multi-step: clean, then model, then chart, then summarise. It plans the steps and works through them in the workbook.

Watch for

Agent Mode changes your file. Work on a copy for anything you cannot afford to rebuild, and read the plan it proposes before letting it run.

Python in Excel

For statistics that native formulas handle badly - regression, clustering, distribution testing. Ask Copilot to write and run it, then to explain what the output means.

PROMPTS WORTH THE EXTRA POWER

"Clean this dataset, build a monthly summary by region, chart the trend and write a short commentary. Show me your plan before you start."

"Build me a three-statement-style monthly model from this transaction data, with assumptions in labelled cells and every calculation traceable to them."

"Use Python to test whether the difference between these two groups is statistically meaningful, and explain the result to someone who has never heard of a p-value."

"Run a regression of Revenue against Marketing Spend and Headcount, and tell me in plain English how much each one appears to matter and how confident I should be."

9 Verify before you trust

Five checks, under two minutes, every time it matters

A wrong number in a spreadsheet travels further than a wrong sentence in a document. This is the routine that keeps Copilot useful and keeps you credible.

1 Ask for the formula, always

"Tell me the formula you used" turns an answer into something auditable. If Copilot cannot state it, you cannot defend the number.

2 Check the row count

Ask how many rows were included. Filters, blanks and header confusion silently drop data, and the total still looks plausible.

3 Test one row by hand

Pick a row you know and check it manually. One correct row is weak evidence; one wrong row is conclusive.

4 Probe the edges

"What did you do with blanks, zeros, negatives and cancelled rows?" This is where the quiet errors live.

5 Re-run before you send

Model outputs are not guaranteed to be identical between runs. For anything that leaves your screen, convert the result into a real formula or a pasted value.

WHAT CHANGED IN SEPTEMBER 2026

The =COPILOT worksheet function - the in-cell formula that let you write a prompt and get an AI answer in the grid - was retired on 14 September 2026. It never left preview, and values already calculated remain in workbooks as cached results. Everything it did (summarising, classifying, generating and looking things up) is available in the Copilot pane instead. If you have inherited a workbook that uses it, replace those cells before the cached values mislead someone. Many cheatsheets still in circulation were written before this and still teach the function.

TROUBLESHOOTING

Symptom	Almost always the cause
Copilot is greyed out or missing	No Copilot licence, or the file is not on OneDrive or SharePoint with AutoSave on.
"I can't work with this data"	Merged cells, multiple header rows, or a range that was never formatted as a table. Ctrl+T usually fixes it in one keystroke.
Totals are wrong but look right	Numbers stored as text are skipped by aggregation. Check alignment: right-aligned is numeric, left-aligned is text.
It analyses the wrong columns	Duplicate or blank header names. Rename to something short and unique.
Dates behave strangely	Mixed regional formats in one column. Standardise the column first, then analyse.
Different answer on a re-run	Expected behaviour for generative output. Convert anything you depend on into a formula or a static value.

QUICK REFERENCE

Excel Formula & Analysis Cheatsheet

BEFORE YOU PROMPT

- Ctrl+T the range, one clean header row, no merged cells
- Saved to OneDrive or SharePoint, AutoSave on
- Numbers right-aligned, dates in one format

THE PATTERN

- Location + Action + Rule + Output
- "In the Sales table, add a column called X that... treat blanks as... and tell me the formula"

WORKHORSE PROMPTS

- Describe this dataset and flag anything unreliable
- Find every data quality problem, but do not fix anything yet
- Add a column that [logic], and tell me the formula you used
- Explain what the formula in [cell] does, step by step
- Break the change in [metric] down by [dimension] and tell me where most of it came from
- Forecast [metric] for the next [n] periods and state the method and assumptions
- Chart [metric] so that [the point] is obvious

BEFORE YOU TRUST IT

- How many rows did you include?
- What did you do with blanks, zeros and negatives?
- Tell me the formula, then check one row by hand

Copilot has not made spreadsheet skill obsolete - it has moved it. The scarce skill is no longer remembering the syntax of an index-match. It is knowing which question to ask, and being able to tell when the answer is wrong.